

2. Show how a bottom up parser with the given grammar rules will process the following input string. digit - id + (digit) (12)

$E \rightarrow E + T \mid E - T \mid E * T \mid E / T \mid T$

$T \rightarrow F \mid C$

$F \rightarrow (E) \mid \text{digit} \mid \text{id}$

$C \rightarrow \text{digit} - \text{id}$

Stack	Input Buffer	Action
\$	digit - id + (digit) \$	Shift
\$ digit	- id + (digit) \$	Reduce by $F \rightarrow \text{digit}$
\$ F	- id + (digit) \$	Reduce by $T \rightarrow F$
\$ T	- id + (digit) \$	Reduce by $E \rightarrow T$
\$ E	- id + (digit) \$	Shift
\$ E -	id + (digit) \$	Shift
\$ E - id	+ (digit) \$	Reduce by $F \rightarrow \text{id}$
\$ E - F	+ (digit) \$	Reduce by $T \rightarrow F$
\$ E - T	+ (digit) \$	Reduce by $E \rightarrow E - T$
\$ E	+ (digit) \$	Shift
\$ E +	(digit) \$	Shift
\$ E + (	digit) \$	Shift
\$ E + (digit	) \$	Reduce by $F \rightarrow \text{digit}$
\$ E + (F	) \$	Reduce by $T \rightarrow F$
\$ E + (T	) \$	Reduce by $E \rightarrow T$
\$ E + (E	) \$	Shift
\$ E + (E)	\$	Reduce by $F \rightarrow (E)$
\$ E + F	\$	Reduce by $T \rightarrow F$
\$ E + T	\$	Reduce by $E \rightarrow E + T$
\$ E	\$	accept.

2. Show how a bottom up parser with the given grammar rules will process the following input string. (digit / (digit)) (12)

$E \rightarrow E + T \mid E - T \mid E * T \mid E / T \mid T$

$T \rightarrow F \mid C$

$F \rightarrow (E) \mid \text{digit} \mid \text{id}$

$C \rightarrow \text{digit} - \text{id}$

Stack	Input Buffer	Action
\$	(digit / (digit))\$	Shift
\$(	digit / (digit))\$	Shift
\$(digit	/ (digit))\$	Reduce by $F \rightarrow \text{digit}$
\$(F	/ (digit))\$	Reduce by $T \rightarrow F$
\$(T	/ (digit))\$	Reduce by $E \rightarrow T$
\$(E	/ (digit))\$	Shift
\$(E/	(digit))\$	Shift
\$(E/(	digit))\$	Shift
\$(E/(digit	))\$	Reduce by $F \rightarrow \text{digit}$
\$(E/(F	))\$	Reduce by $T \rightarrow F$
\$(E/(T	))\$	Reduce by $E \rightarrow T$
\$(E/(E	))\$	Shift
\$(E/(E)	)\$	Reduce by $F \rightarrow (E)$
\$(E/F	)\$	Reduce by $T \rightarrow F$
\$(E/T	)\$	Reduce by $E \rightarrow E/T$
\$(E	)\$	Shift
\$(E)	\$	Reduce by $F \rightarrow (E)$
\$(F	\$	Reduce by $T \rightarrow F$
\$T	\$	Reduce by $E \rightarrow T$
\$E	\$	accept

2. Show how a bottom up parser with the given grammar rules will process the following input string. $id + digit / (id)$ (12)

$E \rightarrow E + T \mid E - T \mid E * T \mid E / T \mid T$

$T \rightarrow F \mid C$

$F \rightarrow (E) \mid digit \mid id$

$C \rightarrow digit - id$

Stack	Input Buffer	Action
\$	$id + digit / (id) \$$	Shift
$\$ id$	$+ digit / (id) \$$	Reduce by $F \rightarrow id$
$\$ F$	$+ digit / (id) \$$	Reduce by $T \rightarrow F$
$\$ T$	$+ digit / (id) \$$	Reduce by $E \rightarrow T$
$\$ E$	$+ digit / (id) \$$	Shift
$\$ E +$	$digit / (id) \$$	Shift
$\$ E + digit$	$/ (id) \$$	Reduce by $F \rightarrow digit$
$\$ E + F$	$/ (id) \$$	Reduce by $T \rightarrow F$
$\$ E + T$	$/ (id) \$$	Reduce by $E \rightarrow E + T$
$\$ E$	$/ (id) \$$	Shift
$\$ E /$	$(id) \$$	Shift
$\$ E / ($	$id) \$$	Shift
$\$ E / (id$	$) \$$	Reduce by $F \rightarrow id$
$\$ E / (F$	$) \$$	Reduce by $T \rightarrow F$
$\$ E / (T$	$) \$$	Reduce by $E \rightarrow T$
$\$ E / (E$	$) \$$	Shift
$\$ E / (E)$	$\$$	Reduce by $F \rightarrow (E)$
$\$ E / F$	$\$$	Reduce by $T \rightarrow F$
$\$ E / T$	$\$$	Reduce by $E \rightarrow E / T$
$\$ E$	$\$$	accept